Geometry & Measurement

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Angle facts**

Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

Angles on a straight line sum to $180°$: $3x + 2x + 30 = 180 \Rightarrow 5x = 150 \Rightarrow x = 30$.

2 **[Foundational]** • **Target: Angle facts**

Find the angle vertically opposite a $65°$ angle.

Vertically opposite angles are equal, so the angle is $65°$.

3 **[Foundational]** • **Target: Angle facts**

Three angles around a point measure $110°$, $95°$, and x. Find x.

Angles around a point sum to $360°$: $x = 360 - 110 - 95 = 155°$.

4 **[Foundational]** • **Target: Angle facts**

Two angles are complementary (sum to $90°$). One measures $37°$. Find the other.

$90 - 37 = 53°$.

5 **[Foundational]** • **Target: Parallel lines and transversals**

Two parallel lines are cut by a transversal, creating an angle of $70°$. Find its corresponding angle.

Corresponding angles are equal, so the angle is $70°$.

6 **[Foundational]** • **Target: Parallel lines and transversals**

In the same setup, find the co-interior (allied) angle to a $70°$ angle.

Co-interior angles sum to $180°$: $180 - 70 = 110°$.

7 **[Foundational]** • **Target: Parallel lines and transversals**

Find the alternate interior angle to a $55°$ angle.

Alternate interior angles are equal, so the angle is $55°$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Parallel lines and transversals

Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

Co-interior angles sum to $180°$: $4x + 2x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.

9 [Intermediate] • Target: Triangle angle sum

Find the third angle of the triangle shown, given the two marked angles.

$180 - 40 - 75 = 65°$.

10 [Intermediate] • Target: Triangle angle sum

An isosceles triangle has two equal angles of $72°$ each. Find the third angle.

$180 - 72 - 72 = 36°$.

11 [Intermediate] • Target: Triangle angle sum

In a right triangle, one of the non-right angles is $35°$. Find the other non-right angle.

$180 - 90 - 35 = 55°$.

12 [Intermediate] • Target: Triangle angle sum

A triangle has three equal angles. Find each angle, and name this type of triangle.

$180 \div 3 = 60°$ each — this is an equilateral (equiangular) triangle.

13 [Intermediate] • Target: Angle sums of polygons

Find the sum of the interior angles of a hexagon (6 sides).

Sum $= (n - 2) \times 180° = (6 - 2) \times 180° = 720°$.

14 [Intermediate] • Target: Angle sums of polygons

Find the sum of the interior angles of a pentagon (5 sides).

Sum $= (5 - 2) \times 180° = 540°$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Angle sums of polygons • Cross-Lesson: 1.01 Arithmetic & Number Sense

A regular polygon has an interior angle sum of $1{,}080°$. How many sides does it have?

$(n - 2) \times 180 = 1{,}080 \Rightarrow n - 2 = 6 \Rightarrow n = 8$ sides (a regular octagon).

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16 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the hypotenuse of the right triangle shown.

$c = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ cm.

17 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 3.01 Algebra Essentials

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

$\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

18 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 5.01 Quantitative Comparisons

A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

$\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ m.

19 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 6.01 Sequences & Patterns

A right triangle has legs 9 and 12. Find the hypotenuse.

$\sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

20 [Challenging] • Target: Perimeter and area of rectangles and squares • Cross-Lesson: 7.01 Data Analysis & Probability

A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

Perimeter $= 2(12 + 7) = 38$ cm. Area $= 12 \times 7 = 84$ cm^2.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Angle facts

Two angles are supplementary. One is three times the other. Find the smaller angle.

a. 60°

b. 30°

c. 135°

d. 45° ✓

$x + 3x = 180 \Rightarrow x = 45^\circ$.

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22 [Foundational] • Target: MCQ — Angle facts

An angle is four times its complement. Find the angle.

a. 54°

b. 18°

c. 72° ✓

d. 90°

$x = 4(90 - x) \Rightarrow 5x = 360 \Rightarrow x = 72^\circ$.

23 [Foundational] • Target: MCQ — Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x.

a. 10

b. 8

c. 30

d. 20 ✓

Corresponding angles are equal: $3x + 15 = 5x - 25 \Rightarrow x = 20$.

24 [Foundational] • Target: MCQ — Parallel lines and transversals

Two parallel lines cut by a transversal create co-interior angles of $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x.

a. 30

b. 40 ✓

c. 50

d. 34

Co-interior angles sum to 180°: $2x + 10 + 3x - 30 = 180 \Rightarrow 5x = 200 \Rightarrow x = 40$.

25 [Foundational] • Target: MCQ — Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle is 63°. Find its alternate exterior angle.

a. 153°

b. 27°

c. 63° ✓

d. 117°

Alternate exterior angles are equal, so the answer is 63°.

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26 [Foundational] • Target: MCQ — Parallel lines and transversals

Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180°. Find the larger angle.

a. 110° ✓

b. 70°

c. 140°

d. 90°

$L + S = 180, L - S = 40 \Rightarrow L = 110^\circ$.

27 [Foundational] • Target: MCQ — Triangle angle sum

Find the third angle of the triangle shown.

a. 116°

b. 64°

c. 54°

d. 44° ✓

$180 - 64 - 72 = 44^\circ$.

28 [Foundational] • Target: MCQ — Triangle angle sum

The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.

a. 55°

b. 70°

c. 40° ✓

d. 110°

$180 - 70 - 70 = 40^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Triangle angle sum

A triangle has three angles in the ratio $2 : 3 : 4$. Find the largest angle.

a. 90°

b. 100°

c. 80° ✓

d. 60°

9 parts $= 180^\circ$, so 1 part $= 20^\circ$; largest $= 4 \times 20 = 80^\circ$.

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30 [Intermediate] • Target: MCQ — Triangle angle sum

In a right triangle, one non-right angle exceeds the other by 26°. Find the larger non-right angle.

a. 32°

b. 58° ✓

c. 64°

d. 48°

$x + (x + 26) = 90 \Rightarrow x = 32$; larger $= 58^\circ$.

31 [Intermediate] • Target: MCQ — The Pythagorean theorem

Find the missing leg of the right triangle shown.

a. 26

b. 18

c. 24 ✓

d. 20

$\sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$.

32 [Intermediate] • Target: MCQ — The Pythagorean theorem

A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.

a. 30 ✓

b. 36

c. 42

d. 26

$\sqrt{24^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$ m.

33 [Intermediate] • Target: MCQ — The Pythagorean theorem

A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?

a. Only if it is also isosceles

b. **Yes — a right triangle** ✓

c. Cannot be determined

d. No — not a right triangle

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so yes, it is a right triangle.

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34 [Intermediate] • Target: MCQ — The Pythagorean theorem

A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?

a. 13

b. 10

c. 12 ✓

d. 18

$\sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144} = 12$ m.

35 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the rectangle shown.

a. $P = 48$ **cm,** $A = 135$ **cm**^{2} ✓

b. $P = 48$ cm, $A = 24$ cm^2

c. $P = 135$ cm, $A = 48$ cm^2

d. $P = 24$ cm, $A = 135$ cm^2

$P = 2(15 + 9) = 48$ cm; $A = 15 \times 9 = 135$ cm^2.

36 [Intermediate] • Target: MCQ — Perimeter and area of rectangles and squares

Find the perimeter and area of the square shown.

a. $P = 44$ cm, $A = 110$ cm^2

b. $P = 44$ **cm,** $A = 121$ **cm**^{2} ✓

c. $P = 22$ cm, $A = 121$ cm^2

d. $P = 121$ cm, $A = 44$ cm^2

$P = 4 \times 11 = 44$ cm; $A = 11^2 = 121$ cm^2.

37 [Intermediate] • Target: MCQ — Area of triangles and parallelograms

The triangle shown has base 16 cm and height 10 cm. Find its area.

a. 160

b. 26

c. 80 ✓

d. 40

$A = \frac{1}{2}(16)(10) = 80$ cm^2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Area of triangles and parallelograms • Cross-Lesson: 5.01 Quantitative Comparisons

The parallelogram shown has base 18 cm and height 7 cm. Find its area.

a. 252

b. 126 ✓

c. 63

d. 25

$A = 18 \times 7 = 126$ cm^2.

39 [Challenging] • Target: MCQ — Area of triangles and parallelograms • Cross-Lesson: 6.01 Sequences & Patterns

A triangle has area 84 cm^2 and height 12 cm. Find its base.

a. 14 ✓

b. 1,008

c. 28

d. 7

$b = \frac{2(84)}{12} = 14$ cm.

40 [Challenging] • Target: MCQ — Area of triangles and parallelograms • Cross-Lesson: 7.01 Data Analysis & Probability

A parallelogram has area 150 cm^2 and base 15 cm. Find its height.

a. 20

b. 10 ✓

c. 2,250

d. 5

$h = \frac{150}{15} = 10$ cm.

41 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the circumference of the circle shown, using $\pi = \frac{22}{7}$.

a. 44

b. 616

c. 88 ✓

d. 176

$C = 2 \times \frac{22}{7} \times 14 = 88$ cm.

42 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find the area of the circle shown, using $\pi = \frac{22}{7}$.

a. 132

b. 2,772

c. 693

d. 1,386 ✓

$A = \frac{22}{7} \times 21^2 = 1{,}386$ cm^2.

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43 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the circumference of the circle shown, in terms of π.

a. 48π cm

b. 24π **cm** ✓

c. 12π cm

d. 144π cm

$C = 2\pi r = 2\pi(12) = 24\pi$ cm.

44 [Challenging] • Target: MCQ — Circles: circumference and area • Cross-Lesson: 3.01 Algebra Essentials

A circle has area 144π cm^2. Find its radius.

a. 6

b. 24

c. 12 ✓

d. 72

$r^2 = 144 \Rightarrow r = 12$ cm.

45 [Challenging] • Target: MCQ — Composite figures • Cross-Lesson: 5.01 Quantitative Comparisons

Find the area of the L-shaped room shown (all measurements in meters).

a. 70

b. 58 ✓

c. 46

d. 82

Full rectangle $10 \times 7 = 70$ minus the 4×3 notch $= 70 - 12 = 58$ m^2.

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